

OMAR AL-BAKRI

Senior Applied AI & Agentic Systems Engineer | Python + TypeScript | LLM Agents, RAG, Evals & Full-Stack Product Delivery
Thailand-based, currently travelling in Indonesia | Fully remote worldwide | UK work authorisation
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| call-score.com

PROFILE

Senior Applied AI Engineer and product builder focused on turning LLM and agent capabilities into reliable, user-facing systems. Builds production-oriented AI products end to end across Python, TypeScript, full-stack application engineering, agent orchestration, RAG and context pipelines, evaluation, observability, integrations and controlled autonomous execution. Founded and operates CallScore and built thredOS, ruleIQ and Helios from scratch, combining hands-on engineering with product ownership, commercial judgement and 15+ years of financial-services experience.

APPLIED AI & AGENTIC ENGINEERING

- Production agents: Stateful workflows, planning, typed tool use, retries, memory, permissions, human review, failure handling, recovery paths and controlled autonomy.
- Full-stack ownership: Problem framing and architecture through backend, frontend, APIs, integrations, testing, deployment, monitoring, documentation and iteration.
- RAG, context and memory: Source-grounded retrieval, GraphRAG/domain graphs, provenance-aware context assembly, PostgreSQL, Redis and Neo4j state and memory.
- Evaluation and reliability: Structured outputs, schema validation, golden cases, regression tests, trace review, hallucination controls, fallbacks, rate limits, circuit breakers and cost monitoring.
- Core stack: Python, TypeScript, FastAPI, Node.js/Express, Next.js/React, REST APIs, LangGraph/LangChain, Zod/Pydantic, PostgreSQL, Redis, Neo4j, Docker and GitHub Actions.

SELECTED APPLIED AI SYSTEMS

CallScore | Founder | Live Market-Intelligence and Multi-Agent Operating System

- Founded and operates a live public platform converting long-form creator content into auditable market-call records with timestamps, evidence receipts, outcome tracking and transparent scoring.
 - Built a multi-agent operating layer spanning data acquisition, extraction, verification, market-data matching, scoring, connected-app automation, review queues, tests and monitoring.
 - Implemented typed graph state, authority-based routing, dry-run orchestration, fail-closed controls, receipts, fallback routes and evidence-based methodology.
- Stack: Python, TypeScript, Next.js, LangGraph, LangChain Core, Zod, PostgreSQL, Redis, PostHog, Composio MCP, Sentry, Netlify

thredOS | Local-First Workflow OS for Multi-Agent Execution

- Built a desktop-first workflow OS for long-horizon multi-agent work across typed threads, DAG dependencies, tools, models, policies, provenance and resumable execution.
 - Delivered Desktop, Node SDK and Thread-Runner layers with React Flow, CLI handlers, 21 API route groups, model-agnostic routing, permission controls and recovery paths.
- Stack: Bun, TypeScript, Next.js, React Flow, Zustand, TanStack Query, typed APIs, model routing and policy controls

ruleIQ | Agentic Compliance and Governance Platform

- Built a full-stack AI compliance platform with LangGraph agents, an explicit state loop, GraphRAG/Neo4j knowledge layer, structured outputs and hallucination controls.
 - Implemented JWT/RBAC, AES-GCM encryption, Pydantic-style validation, observability-ready logging and 1,884+ backend tests across 56+ API routers and 140+ services.
- Stack: FastAPI, Python, PostgreSQL, Neo4j, Redis, LangGraph, Next.js, Docker, GitHub Actions CI

Helios | AI-Driven Commercial Automation Platform at RTGS.global

- Built and deployed Helios, replacing fragmented manual prospecting workflows and increasing lead conversion by 25%.
- Integrated Explorium, Apollo, HubSpot and Postmark with Electron/React, Node/Express, FastAPI, PostgreSQL, Redis and Neo4j/Graphiti.
- Built human-in-the-loop agent workflows with execution limits, guardrails, circuit breakers, cost caps, cancellation windows and multi-provider routing.

PROFESSIONAL EXPERIENCE

Independent Applied AI Engineer & Product Builder | Fully Remote | 2023-Present

- Builds production-oriented AI products across agent design, architecture, frontend, backend, data/retrieval flows, evaluation, automated testing, deployment, observability and documentation.
- Combines hands-on engineering with product judgement, customer discovery, commercial prioritisation and senior stakeholder communication.

RTGS.global | Director, Partnerships & Client Services; Asia Territory Lead; AI Lead | Oct 2024-Mar 2026

- Built and deployed Helios, increasing lead conversion by 25%.
- Owned AI strategy for commercial execution and worked across Product, Engineering, Compliance and Commercial teams.

Banking Circle | Senior Account Manager | Mar 2023-Sep 2023

- Introduced AI-assisted prospecting, increasing response rates by 20%.
- Led onboarding redesign that reduced onboarding time by 50% and supported 30%+ payment-volume growth.

Convera | Head of Bank Relationship Sales, EMEA; Senior Business Development Manager | Feb 2021-Feb 2023

- Closed a GBP 1M financial-institution partnership and expanded banking relationships contributing to 30% payment-volume growth.

Earlier Career | 2006-2020

Financial services, executive search and advisory, including Bank of America Merrill Lynch, BNP Paribas CIB, Marlin Hawk Group, Hoggett Bowers, Alderbrooke Group and Leathwaite International. Won 10 global clients including Santander and Vanguard, generating GBP 1.53M in fees.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL

Frontend: Next.js, React, Electron, React Flow, Zustand, TanStack Query, Tailwind CSS

Backend: Node.js, Express, FastAPI, REST APIs, webhooks, typed routes, service decomposition

Agents: LangGraph, LangChain, CrewAI/AutoGen patterns, custom multi-agent orchestration, MCP-style tools

State and data: PostgreSQL, Redis, Neo4j/Graphiti, GraphRAG, retrieval pipelines, checkpointing

Evaluation: Golden cases, regression suites, prompt evaluation, structured outputs, validation, trace review

Reliability: Sentry, LangSmith/Langfuse-style traces, rate limits, fallbacks, circuit breakers, latency and cost tracking

LLMs: OpenAI, Anthropic/Claude, Gemini, OpenRouter, model routing, streaming and caching

Domain: Agentic products, developer tooling, fintech/payments, compliance automation, market intelligence and growth systems

EDUCATION

BSc (Hons) Physiology / Pharmacology - Nottingham Trent University